# Package 'betaSandwich'

November 2, 2026

**Title** Robust Confidence Intervals for Standardized Regression Coefficients

**Version** 1.0.9.9000

**Description** Generates robust confidence intervals for standardized regression coefficients using heteroskedasticity-consistent standard errors for models fitted by lm() as described in Dudgeon (2017) <doi:10.1007/s11336-017-9563-z>. The package can also be used to generate confidence intervals for R-squared, adjusted R-squared, and differences of standardized regression coefficients. A description of the package and code examples are presented in Pesigan, Sun, and Cheung (2023) <doi:10.1080/00273171.2023.2201277>.

**URL** https://github.com/jeksterslab/betaSandwich, https://jeksterslab.github.io/betaSandwich/

**BugReports** https://github.com/jeksterslab/betaSandwich/issues

**License** MIT + file LICENSE

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`BetaADF` *Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Using the Asymptotic Distribution-Free Approach*

---

## Description

Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Using the Asymptotic Distribution-Free Approach

## Usage

```
BetaADF(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| `object` | Object of class `lm`. |
| `alpha` | Numeric vector. Significance level $\alpha$. |

## Details

Note that while the calculation in `BetaADF()` is different from `betaDelta::BetaDelta()` with `type = "adf"`, the results are numerically equivalent. `BetaADF()` is appropriate when sample sizes are moderate to large (`n > 250`). `BetaHC()` is recommended in most situations.

## Value

Returns an object of class `betasandwich` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm_process** Processed `lm` object.

**gamma_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Browne, M. W. (1984). Asymptotically distribution-free methods for the analysis of covariance structures. *British Journal of Mathematical and Statistical Psychology*, *37*(1), 62–83. doi:10.1111/j.20448317.1984.tb00789.x

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

## See Also

Other Beta Sandwich Functions: `BetaHC()`, `BetaN()`, `DiffBetaSandwich()`, `RSqBetaSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaADF(object)
# Methods -----------------------------------------------------
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

BetaHC *Estimate Standardized Regression Coefficients and the Corresponding Robust Sampling Covariance Matrix Using the Heteroskedasticity Consistent Approach*

## Description

Estimate Standardized Regression Coefficients and the Corresponding Robust Sampling Covariance Matrix Using the Heteroskedasticity Consistent Approach

## Usage

```
BetaHC(
  object,
  type = "hc3",
  alpha = c(0.05, 0.01, 0.001),
  g1 = 1,
  g2 = 1.5,
  k = 0.7
)
```

## Arguments

| | |
|---|---|
| object | Object of class lm. |
| type | Character string. Correction type. Possible values are "hc0", "hc1", "hc2", "hc3", "hc4", "hc4m", and "hc5". |
| alpha | Numeric vector. Significance level $\alpha$. |
| g1 | Numeric. g1 value for type = "hc4m". |
| g2 | Numeric. g2 value for type = "hc4m". |
| k | Numeric. Constant k for type = "hc5" $0 \leq k \leq 1$. |

## Value

Returns an object of class betasandwich which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm_process** Processed lm object.

**gamma_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaN()`, `DiffBetaSandwich()`, `RSqBetaSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
# Methods -----------------------------------------------------
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

---

`BetaN` *Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Assuming Multivariate Normality*

---

## Description

Estimate Standardized Regression Coefficients and the Corresponding Sampling Covariance Matrix Assuming Multivariate Normality

## Usage

```
BetaN(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| `object` | Object of class `lm`. |
| `alpha` | Numeric vector. Significance level $\alpha$. |

## Details

Note that while the calculation in `BetaN()` is different from `betaDelta::BetaDelta()` with `type = "mvn"`, the results are numerically equivalent. `BetaN()` assumes multivariate normality. `BetaHC()` is recommended in most situations.

## Value

Returns an object of class `betasandwich` which is a list with the following elements:

**call** Function call.

**args** Function arguments.

**lm_process** Processed `lm` object.

**gamma_n** Asymptotic covariance matrix of the sample covariance matrix assuming multivariate normality.

**gamma_hc** Asymptotic covariance matrix HC correction.

**gamma** Asymptotic covariance matrix of the sample covariance matrix.

**acov** Asymptotic covariance matrix of the standardized slopes.

**vcov** Sampling covariance matrix of the standardized slopes.

**est** Vector of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## References

Dudgeon, P. (2017). Some improvements in confidence intervals for standardized regression coefficients. *Psychometrika*, *82*(4), 928–951. doi:10.1007/s113360179563z

Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*. doi:10.1080/00273171.2023.2201277

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `DiffBetaSandwich()`, `RSqBetaSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaN(object)
# Methods -----------------------------------------------------
print(std)
summary(std)
coef(std)
vcov(std)
confint(std, level = 0.95)
```

## coef.betasandwich *Standardized Regression Slopes*

### Description

Standardized Regression Slopes

### Usage

```
## S3 method for class 'betasandwich'
coef(object, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class betasandwich. |
| ... | additional arguments. |

### Value

Returns a vector of standardized regression slopes.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
coef(std)
```

## coef.diffbetasandwich *Differences of Standardized Regression Slopes*

### Description

Differences of Standardized Regression Slopes

### Usage

```
## S3 method for class 'diffbetasandwich'
coef(object, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class diffbetasandwich. |
| ... | additional arguments. |

### Value

Returns a vector of differences of standardized regression slopes.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
coef(diff)
```

---

## coef.rsqbetasandwich *Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)*

---

### Description

Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)

### Usage

```
## S3 method for class 'rsqbetasandwich'
coef(object, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class rsqbetasandwich. |
| ... | additional arguments. |

### Value

Returns a vector of multiple correlation coefficients (R-squared and adjusted R-squared)

### Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
coef(rsq)
```

---

`confint.betasandwich` *Confidence Intervals for Standardized Regression Slopes*

---

## Description

Confidence Intervals for Standardized Regression Slopes

## Usage

```
## S3 method for class 'betasandwich'
confint(object, parm = NULL, level = 0.95, ...)
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`. |
| `parm` | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| `level` | the confidence level required. |
| `...` | additional arguments. |

## Value

Returns a matrix of confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
confint(std, level = 0.95)
```

confint.diffbetasandwich *Confidence Intervals for Differences of Standardized Regression Slopes*

## Description

Confidence Intervals for Differences of Standardized Regression Slopes

## Usage

```
## S3 method for class 'diffbetasandwich'
confint(object, parm = NULL, level = 0.95, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class diffbetasandwich. |
| parm | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level | the confidence level required. |
| ... | additional arguments. |

## Value

Returns a matrix of confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
confint(diff, level = 0.95)
```

confint.rsqbetasandwich *Confidence Intervals for Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)*

## Description

Confidence Intervals for Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)

## Usage

```
## S3 method for class 'rsqbetasandwich'
confint(object, parm = NULL, level = 0.95, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class rsqbetasandwich. |
| parm | a specification of which parameters are to be given confidence intervals, either a vector of numbers or a vector of names. If missing, all parameters are considered. |
| level | the confidence level required. |
| ... | additional arguments. |

## Value

Returns a matrix of confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
confint(rsq, level = 0.95)
```

`DiffBetaSandwich` *Estimate Differences of Standardized Slopes and the Corresponding Sampling Covariance Matrix*

## Description

Estimate Differences of Standardized Slopes and the Corresponding Sampling Covariance Matrix

## Usage

```
DiffBetaSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions. |
| `alpha` | Numeric vector. Significance level $\alpha$. |

## Value

Returns an object of class `diffbetasandwich` which is a list with the following elements:

**call** Function call.

**fit** The argument `object`.

**args** Function arguments.

**vcov** Sampling covariance matrix of differences of standardized slopes.

**est** Vector of differences of standardized slopes.

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `BetaN()`, `RSqBetaSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
# Methods -----------------------------------------------------
print(diff)
summary(diff)
coef(diff)
vcov(diff)
confint(diff, level = 0.95)
```

## nas1982 *1982 National Academy of Sciences Doctoral Programs Data*

### Description

1982 National Academy of Sciences Doctoral Programs Data

### Usage

```
nas1982
```

### Format

Ratings of 46 doctoral programs in psychology in the USA with the following variables:

**QUALITY** Program quality ratings.

**NFACUL** Number of faculty members in the program.

**NGRADS** Number of program graduates.

**PCTSUPP** Percentage of program graduates who received support.

**PCTGRT** Percent of faculty members holding research grants.

**NARTIC** Number of published articles attributed to program faculty member.

**PCTPUB** Percent of faculty with one or more published article.

### References

National Research Council. (1982). *An assessment of research-doctorate programs in the United States: Social and behavioral sciences*. doi:10.17226/9781. Reproduced with permission from the National Academy of Sciences, Courtesy of the National Academies Press, Washington, D.C.

## print.betasandwich *Print Method for an Object of Class* betasandwich

### Description

Print Method for an Object of Class betasandwich

### Usage

```
## S3 method for class 'betasandwich'
print(x, alpha = NULL, digits = 4, ...)
```

### Arguments

| | |
|---|---|
| x | Object of class betasandwich. |
| alpha | Numeric vector. Significance level $\alpha$. If alpha = NULL, use the argument alpha used in x. |
| digits | Digits to print. |
| ... | additional arguments. |

### Value

Prints a matrix of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
print(std)
```

---

## print.diffbetasandwich

*Print Method for an Object of Class* diffbetasandwich

---

### Description

Print Method for an Object of Class diffbetasandwich

### Usage

```
## S3 method for class 'diffbetasandwich'
print(x, alpha = NULL, digits = 4, ...)
```

### Arguments

| | |
|---|---|
| x | Object of class diffbetasandwich. |
| alpha | Numeric vector. Significance level $\alpha$. If alpha = NULL, use the argument alpha used in x. |
| digits | Digits to print. |
| ... | additional arguments. |

## Value

Prints a matrix of differences of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
print(diff)
```

---

# print.rsqbetasandwich *Print Method for an Object of Class* rsqbetasandwich

---

## Description

Print Method for an Object of Class `rsqbetasandwich`

## Usage

```
## S3 method for class 'rsqbetasandwich'
print(x, alpha = NULL, digits = 4, ...)
```

## Arguments

| | |
|---|---|
| `x` | Object of class `rsqbetasandwich`. |
| `alpha` | Numeric vector. Significance level $\alpha$. If `alpha = NULL`, use the argument `alpha` used in `x`. |
| `digits` | Digits to print. |
| `...` | additional arguments. |

## Value

Prints a matrix of multiple correlation coefficients (R-squared and adjusted R-squared), standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
print(rsq)
```

---

`RSqBetaSandwich` *Estimate Multiple Correlation Coefficients (R-squared and adjusted R-squared) and the Corresponding Sampling Covariance Matrix*

---

## Description

Estimate Multiple Correlation Coefficients (R-squared and adjusted R-squared) and the Corresponding Sampling Covariance Matrix

## Usage

```
RSqBetaSandwich(object, alpha = c(0.05, 0.01, 0.001))
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`, that is, the output of the `BetaHC()`, `BetaN()`, or `BetaADF()` functions. |
| `alpha` | Numeric vector. Significance level $\alpha$. |

## Value

Returns an object of class `rsqbetasandwich` which is a list with the following elements:

**call** Function call.

**fit** The argument `object`.

**args** Function arguments.

**vcov** Sampling covariance matrix of multiple correlation coefficients (R-squared and adjusted R-squared).

**est** Vector of multiple correlation coefficients (R-squared and adjusted R-squared).

## Author(s)

Ivan Jacob Agaloos Pesigan

## See Also

Other Beta Sandwich Functions: `BetaADF()`, `BetaHC()`, `BetaN()`, `DiffBetaSandwich()`

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
# Methods ------------------------------------------------------
print(rsq)
summary(rsq)
coef(rsq)
vcov(rsq)
confint(rsq, level = 0.95)
```

---

`summary.betasandwich` *Summary Method for an Object of Class* `betasandwich`

---

## Description

Summary Method for an Object of Class `betasandwich`

## Usage

```
## S3 method for class 'betasandwich'
summary(object, alpha = NULL, digits = 4, ...)
```

## Arguments

| | |
|---|---|
| `object` | Object of class `betasandwich`. |
| `alpha` | Numeric vector. Significance level $\alpha$. If `alpha = NULL`, use the argument `alpha` used in `object`. |
| `digits` | Digits to print. |
| `...` | additional arguments. |

## Value

Returns a matrix of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
summary(std)
```

summary.diffbetasandwich *Summary Method for an Object of Class* diffbetasandwich

## Description

Summary Method for an Object of Class diffbetasandwich

## Usage

```
## S3 method for class 'diffbetasandwich'
summary(object, alpha = NULL, digits = 4, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class diffbetasandwich. |
| alpha | Numeric vector. Significance level $\alpha$. If alpha = NULL, use the argument alpha used in object. |
| digits | Digits to print. |
| ... | additional arguments. |

## Value

Returns a matrix of differences of standardized regression slopes, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
summary(diff)
```

summary.rsqbetasandwich *Summary Method for an Object of Class* rsqbetasandwich

## Description

Summary Method for an Object of Class rsqbetasandwich

## Usage

```
## S3 method for class 'rsqbetasandwich'
summary(object, alpha = NULL, digits = 4, ...)
```

## Arguments

| | |
|---|---|
| object | Object of class rsqbetasandwich. |
| alpha | Numeric vector. Significance level $\alpha$. If alpha = NULL, use the argument alpha used in object. |
| digits | Digits to print. |
| ... | additional arguments. |

## Value

Returns a matrix of multiple correlation coefficients (R-squared and adjusted R-squared), standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

## Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
summary(rsq)
```

## vcov.betasandwich *Sampling Covariance Matrix of the Standardized Regression Slopes*

### Description

Sampling Covariance Matrix of the Standardized Regression Slopes

### Usage

```
## S3 method for class 'betasandwich'
vcov(object, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class betasandwich. |
| ... | additional arguments. |

### Value

Returns a matrix of the variance-covariance matrix of standardized slopes.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
vcov(std)
```

## vcov.diffbetasandwich *Sampling Covariance Matrix of Differences of Standardized Regression Slopes*

### Description

Sampling Covariance Matrix of Differences of Standardized Regression Slopes

### Usage

```
## S3 method for class 'diffbetasandwich'
vcov(object, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class diffbetasandwich. |
| ... | additional arguments. |

### Value

Returns a matrix of the variance-covariance matrix of differences of standardized regression slopes.

### Author(s)

Ivan Jacob Agaloos Pesigan

### Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
diff <- DiffBetaSandwich(std)
vcov(diff)
```

---

## vcov.rsqbetasandwich *Sampling Covariance Matrix of Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)*

---

### Description

Sampling Covariance Matrix of Multiple Correlation Coefficients (R-Squared and Adjusted R-Squared)

### Usage

```
## S3 method for class 'rsqbetasandwich'
vcov(object, ...)
```

### Arguments

| | |
|---|---|
| object | Object of class rsqbetasandwich. |
| ... | additional arguments. |

### Value

Returns a matrix of the variance-covariance matrix of multiple correlation coefficients (R-squared and adjusted R-squared).

### Author(s)

Ivan Jacob Agaloos Pesigan

## Examples

```
object <- lm(QUALITY ~ NARTIC + PCTGRT + PCTSUPP, data = nas1982)
std <- BetaHC(object)
rsq <- RSqBetaSandwich(std)
vcov(rsq)
```

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